

Lantern Core MK II - COMET LEAP Arc Reactor Housing Report

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Repo: alex-place/lantern-os
Mode: procurement/RFQ package, safety-gated, external wearable only

BOUNDARY

ChatGPT cannot order parts, pay suppliers, arrange China assembly, receive goods, assemble hardware, or ship hardware.
This report is the procurement and manufacturing handoff: BOM tiers, safety gates, RFQ language, prototype sequence, and COMET LEAP sprint.

PRODUCT FRAME

Lantern Core MK II is a safe external cyborg wearable: a low-voltage chest lantern core with tiny haptic feedback, passive identity, and future BLE/Wi-Fi/Starlink path through a phone or router.

- It is not:
- an implant;
 - a medical device;
 - life support;
 - a diagnostic system;
 - a tool for high-power stimulation;
 - a device to charge while worn.

COMET LEAP DECISION

- Build MK0.5 first.
- Do not jump straight to mass assembly or body-critical features. The first safe win is a low-voltage external chest puck with:
- diffused lantern glow;
 - tiny haptic actuator like a soft DualShock pulse;
 - tactile button;
 - passive NFC / Java Card identity layer;
 - soft skin-contact shell;
 - removable protected power;
 - no Wi-Fi required for the first comfort proof.

RESEARCH ANCHORS

- ESP32-S3 is suitable for a later Wi-Fi/BLE version because it supports 2.4 GHz Wi-Fi, Bluetooth 5 LE, GPIOs, touch/PWM style peripherals, security features, and low-power modes.
- DRV2605L-style haptic controllers are suitable for ERM/LRA haptic prototyping.
- Lithium-ion power banks have real recall/fire/burn history, so battery placement and charging rules are a first-class safety gate.
- Any sold Wi-Fi/BLE product needs compliance planning and preferably pre-certified radio modules.
- Java Card / NFC is plausible as a passive/offline identity layer, but private secrets need encryption and careful access control.

12 SAFE NOW LINES

#	Line	Meaning	Allowed now
1	Soft skin shell	TPU/silicone/foam contact surface	yes
2	Glow ring	Diffused LED ring or COB LED	yes

#	Line	Meaning	Allowed now
3	Tiny haptic	LRA or small ERM vibration motor	yes
4	One button	Large tactile input	yes
5	Passive NFC	NFC tag or Java Card carried in housing	yes
6	Offline identity	tap-to-read profile/emergency card	yes
7	USB power	external certified USB power source	yes
8	Quick release	strap/magnet/snap breakaway	yes
9	Over-shirt test	comfort test before skin-contact	yes
10	No charging worn	remove to charge	yes
11	No medical claims	wellness/accessibility/status only	yes
12	Manual parent/operator mode	no hidden actions	yes

12 SOON-TO-BE SAFE LINES

#	Line	Unlock condition
1	BLE connection	MK0.5 comfort passes; use phone/local only
2	Wi-Fi burst mode	firmware limits radio use; no high-power antenna
3	Lantern OS status feed	local dashboard endpoint exists
4	Haptic pattern language	tested with wearer comfort limits
5	Chest contact sensor	detect seating only; no diagnosis
6	IMU gesture input	no emergency automation without confirmation
7	Prosthetic strap mount	mechanical comfort tested
8	Parent/caregiver alert button	manual press only
9	Starlink via router	connect to Wi-Fi router; no satellite hardware on body
10	Encrypted NFC identity	do not store sensitive secrets in plain text
11	Local AI macro relay	no password/payment/account actions
12	Small vendor assembly	DFM, safety review, and prototype burn-in pass

12 FUTURE HELD LINES

#	Future line	Held until
1	Medical monitoring mode	clinician/regulatory review
2	Implant-adjacent use	doctor clearance, especially if pacemaker/monitor appears
3	Always-on microphone	privacy model, explicit consent
4	Always-on camera	privacy model, explicit consent

#	Future line	Held until
5	Electrical stimulation	medical/professional review
6	Charging while worn	formal thermal/safety certification
7	Autonomous emergency calls	false-positive safety design
8	High-power radio	compliance and exposure review
9	Prosthetic control output	fail-safe engineering and clinician/prosthetist review
10	Public retail sale	FCC/CE/UKCA, battery transport, labeling, QA
11	Cloud health records	legal/privacy architecture
12	Mass China assembly	pilot units pass burn, skin, drop, RF, and comfort tests

PROTOTYPE BOM TIERS

Module	Prototype part class	Why	Safety note
Brain	none for MK0.5; ESP32-S3 dev board for MK1	Wi-Fi/BLE later	radio off for first comfort test
Haptic	LRA coin actuator + haptic driver	clean taps/pulses	limit duty cycle and intensity
Alt haptic	small ERM coin motor	DualShock-like buzz	use soft mount, low duty
Light	diffused LED ring or small RGB LED board	lantern glow	current limit; heat check
Identity	NFC tag or Java Card/contactless card	offline tap identity	no private secrets in plain text
Housing	3D-printed shell + silicone/foam back	skin comfort	round edges; cleanable surface
Mount	elastic harness / chest strap / breakaway clip	wearable stability	quick release
Power	certified USB power bank or protected battery module	safe power	do not charge while worn
Protection	inline fuse/PTC, switch, strain relief	failure reduction	required before body tests
Firmware	simple pulse mode -> BLE/Wi-Fi later	staged complexity	no cloud dependency first
Packaging	labeled prototype case + warning card	safe handling	not medical / remove if uncomfortable

CHINA ASSEMBLY STRATEGY

Do not send a vague arc reactor idea to a factory. Send a controlled RFQ package:

- CAD/STL files;
- PCB Gerbers if a custom PCB exists;
- BOM and CPL if using PCB assembly;
- firmware boundary;
- material requirements;
- safety requirements;
- test checklist;
- acceptance criteria.

First China order should be 3-10 non-sale engineering samples, not a consumer product batch.

Candidate path:

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local maker prototype
-> PCB design
-> 3D printed or silicone prototype
-> DFM review
-> PCB assembly house
-> enclosure vendor
-> final assembly house
-> engineering sample burn-in
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VENDOR RFQ MESSAGE

Hello - I need a small external wearable electronics prototype assembled. It is not an implant and not a medical device. It is a low-voltage chest-worn LED/haptic status puck. First batch: 3-10 engineering samples. Required: smooth rounded enclosure, soft skin-contact backing, LED diffuser, small haptic actuator, protected power input, no charging while worn, and optional ESP32-S3 BLE/Wi-Fi module in the later version. Please quote PCB assembly, enclosure printing/molding options, final assembly, and basic functional test. Provide material certifications and confirm no sharp edges, no exposed wiring, no unprotected Li-ion pouch cell, and no high-temperature surface under normal operation.

TEST PLAN

Gate	Test	Pass condition
Bench electrical	run 30 minutes on table	no heat, no reset, no odor
Haptic comfort	10 patterns at low power	comfortable, not startling/painful
Skin/backing	10 minute contact test	no irritation/pressure mark
Strap	bend, sit, breathe, remove	quick release works
Heat	measure shell temp after LED/haptic	warm is fail; skin-safe cool only
Power	disconnect/strain test	no exposed wire; fuse/protection works
NFC	tap with phone	reads intended public info only
BLE/Wi-Fi	MK1 only	connects locally; radio can be disabled
Starlink path	MK2 only	works through router/phone; no satellite hardware on body
Accessibility	wearer can trigger button	usable without fine dexterity
Stop rule	remove immediately if discomfort	documented and followed

11-DAY COMET LEAP SPRINT

Day	Action	Output
0	Freeze MK0.5 spec	this report + BOM
1	Select body location and shell shape	round/lantern gem/flat badge decision
2	Order maker parts manually	cart + receipt, not automated by AI
3	Bench LED/haptic wiring	glow + pulse demo
4	Build soft dummy housing	comfort-only shell
5	Install electronics into shell	no exposed conductors
6	Over-shirt wear test	comfort notes
7	Skin-contact short test	comfort/skin notes
8	NFC identity card test	phone tap demo

Day	Action	Output
9	Haptic language test	12 patterns scored
10	Photos/video proof	Movie 2 public/private proof
11	Decide MK1 upgrade	ESP32-S3 + BLE/Wi-Fi plan

KEY MISSING QUESTIONS

- 1. Do you expect an implanted device soon: pacemaker, loop recorder, neurostimulator, insulin pump, cochlear implant, implanted monitor, or other device?
- 2. Will the reactor sit directly on sternum skin, over shirt, shoulder strap, prosthetic strap, or wheelchair/cane/bag mount?
- 3. Do you want the first shell to be round Arc Reactor, lantern gem, or flat badge?
- 4. Do you want buzz/rumble ERM or precise taps LRA as the main sensation?
- 5. What is the maximum budget for MK0.5 parts: \$50, \$100, \$200, or more?
- 6. Do you want no-radio MK0.5 first, or ESP32-S3 from the beginning?
- 7. Should the NFC profile contain only a public message, or an emergency-contact page?
- 8. What disability/prosthetic need comes first: fatigue cueing, hand-pain macro button, emergency marker, low-vision haptics, mobility mount, or identity/medical info card?

DONE DEFINITION

The next state is complete when:

- MK0.5 parts are selected by a human operator;
- battery and charging rules are written on the device card;
- a soft dummy housing is tested for comfort;
- haptic intensity is tested at low power;
- no radio or vendor assembly is added until comfort proof passes.